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Multimodal AI for detecting deceptive design patterns in web applications

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Proposal and design review | Model training and user validation remain planned

Problem and intended users

01 Developers, designers and QA reviewers need evidence they can inspect before changing an interface.

02 Research supports the problem. Demand for this specific workflow still needs 3–5 genuine user discussions.

03 The output is a suspected cue for human review. It does not establish deceptive intent.

Scope and research question

01 English-language public web pages, one desktop viewport at a time.

02 Four labels: confirmshaming, misleading choice wording, visual interference and urgency cues.

03 Does image, text and DOM fusion improve macro-F1 on websites excluded from training?

04 Login flows, transactions and multi-step cancellation patterns are outside scope.

Research informing the design

01 12 research sources, including 11 identified conference or journal publications and one labelled preprint.

02 Definitions and ontology guide labels. AidUI and UIGuard motivate evidence localisation and UI features.

03 DPGuard establishes prior multimodal work. The contribution is a bounded web workflow and reproducible comparison.

Existing systems

01 Mathur crawler: shopping-web collection and expert analysis.

02 AidUI: screenshot cues and localisation. UIGuard: mobile UI property checks.

03 DPGuard: multimodal large-model reasoning for web and mobile UIs.

04 Four systems are compared across eight criteria in the report. Published scores are not directly ranked.

Technology choices

01 FastAPI over Django or Express for direct Python inference integration.

02 SQLite over PostgreSQL or MongoDB for the single-machine pilot. Revisit shared storage for concurrent deployment.

03 Playwright over Selenium or Puppeteer for capture in an isolated Python worker.

04 React and TypeScript are planned. The supplied demonstration is dependency-free HTML.

Requirements and acceptance

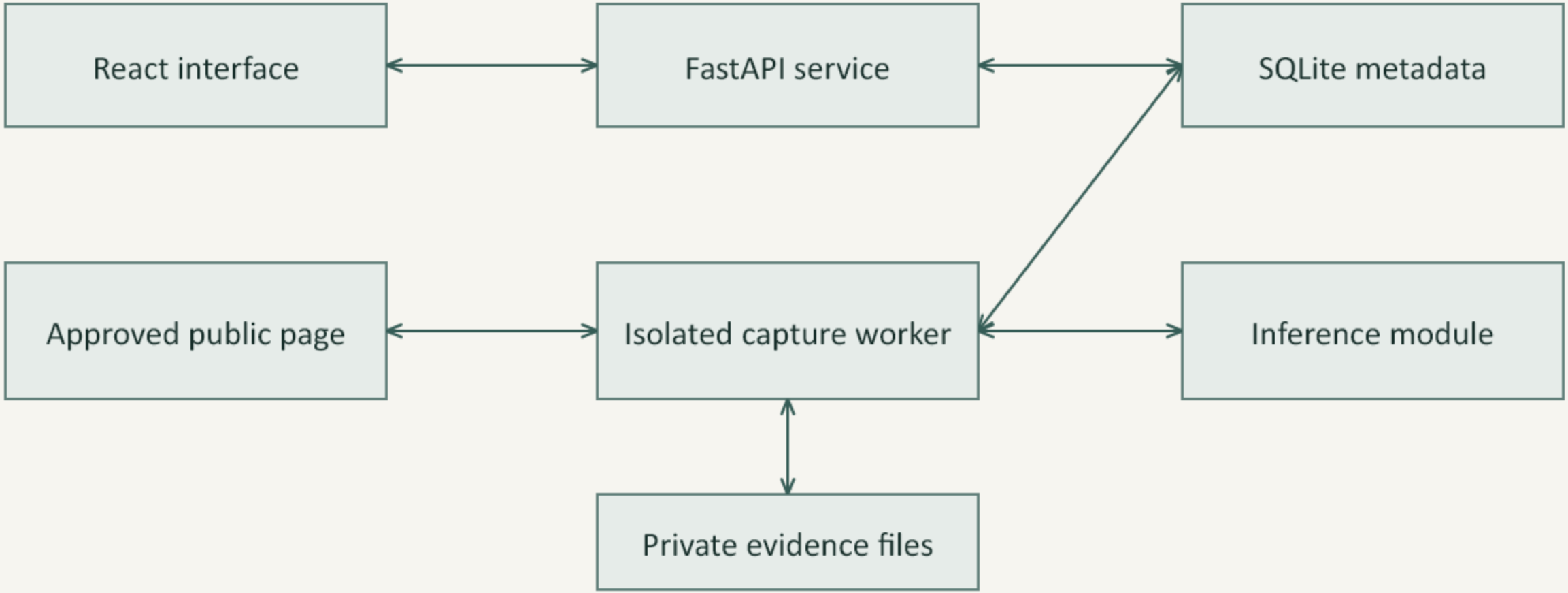
01 Validate permitted targets and show scan progress or explicit failures.

02 Display region evidence, model version and supported labels. Preserve human review separately.

03 Export reviewed JSON. Record annotation provenance and reproducible experiments.

04 Security, runtime, data minimisation and keyboard access have defined checks.

Proposed component architecture



The worker claims persistent jobs and stores results.

Dataset and annotation

01 Target 400–600 regions from at least 30 domains if feasible. Begin with a 50-region pilot.

02 Include benign examples and overlapping labels. Zaki independently reviews at least 20% of Akbar’s annotations.

03 Deduplicate, group by website and shared template, then freeze train, validation and test manifests.

04 These quantities are planned. No completed dataset is claimed.

Evaluation and success criteria

01 Compare text-only, image-only and multimodal models on identical website-separated data.

02 Report per-label precision, recall and F1, macro-F1, benign false positives and runtime.

03 Tune on validation data only. A negative multimodal result remains a valid finding.

04 Proposed targets: macro-F1 ≥ 0.70 and benign false-positive rate ≤ 0.15. No measured results yet.

Demonstration walkthrough

01 Open prototype/index.html locally and choose the fictional shop.

02 Inspect highlighted evidence and an illustrative score. Confirm, dismiss or leave a finding uncertain.

03 Add a note, filter decisions and export the reviewed JSON.

04 Also show neutral and capture-failure examples. The demonstration never fetches a URL or runs AI.

Individual work and milestones

01 Zaki owns capture, API, interface, integration and system testing. Akbar owns data, models and evaluation.

02 Weeks 1–4: capture spike, annotation pilot, requirements and dataset partitions.

03 Weeks 5–8: model comparisons and integration. Weeks 9–12: evaluation, documentation and rehearsal.

04 Proposed window: 26 October–17 January. Confirm the university’s actual final date.

Risks and open evidence

01 Dataset ambiguity and leakage: pilot labels, independent review and grouped splits.

02 Unsafe requests and private data: restricted isolated capture and private evidence storage.

03 Still required: genuine user discussions, supervisor comments and a real assigned YouTrack project.

04 Peer-review the submission, verify access and tag the accepted commit as pre-defense-1.